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# Huawei 50 kW On-Grid Inverter Comparison

SUN2000-50KTL-M3 vs SUN2000-50K-MC0, matched with LONGi Hi-MO X10 Anti-Glare LR8-66HVD-650M (650 Wp, bifacial)

## Summary


- The **50K-MC0** is Huawei's newer successor to the 50KTL-M3 (datasheet dated Oct 2025). Note the model name ends in MC**0** (zero).
- At a 400 V grid both have the **same peak efficiency (98.5%)**; the MC0 is slightly better on European (weighted) efficiency: **98.2% vs 98.0%**.
- The bigger difference is **input current**: the MC0 accepts 23 A per input and up to 33 A / 44 A on MPPT 2 and 3, which suits large bifacial panels such as the LONGi 650 W anti-glare module.
- Both inverters accept **16 to 20 panels per string**. The MC0 is the better match; the M3 is acceptable on rooftops with slight clipping on MPPTs carrying two strings.


## 1. Inverter specification comparison

| Specification | SUN2000-50KTL-M3 | SUN2000-50K-MC0 |
|---|---|---|
| Max. efficiency | 98.5% | 98.5% @400 V (98.7% @480 V) |
| European efficiency | 98.0% | **98.2%** |
| Start voltage | 200 V | **180 V** |
| MPPT operating range | 200 – 1,000 V | **180 – 1,000 V** |
| Max. DC input voltage | 1,100 V | 1,100 V |
| Max. current per input | 20 A | **23 A** |
| Max. current per MPPT | 30 A (all four) | 30 / **33** / **33** / 30 A |
| Max. short-circuit current per MPPT | 40 A | 40 / **44** / **44** / 40 A |
| Inputs / MPP trackers | 8 / 4 | 8 / 4 |
| Rated / max. AC power | 50 kW / 55 kVA | 50 kW / 55 kVA |
| Rated AC voltage | 400 / 480 Vac | **380** / 400 / 480 Vac |
| Rated output current @400 V | 72.2 A | 72.2 A |
| Protections | AFCI, PID recovery, RCMU, Type II SPD (DC & AC) | AFCI, PID recovery, RCMU, Type II SPD (DC & AC) |

**What it means in practice**

- **Efficiency:** the 0.2-point European-efficiency gain gives an estimated ~150–170 kWh/year more on a ~60 kWp array in Chiang Rai (about 700 THB/year). Real, but small.
- **Input current:** the main reason to choose the MC0 with modern 600 W+ bifacial panels, where rear-side gain pushes string current above 20 A.
- **Lower start voltage:** slightly earlier wake-up and later shutdown, a little better on cloudy mornings.
- **380 Vac rating:** officially covers sites where the PEA voltage runs low.

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## 2. Panel data: LONGi Hi-MO X10 LR8-66HVD-650M (Anti-Glare)

| Parameter (STC) | Value | Parameter | Value |
|---|---|---|---|
| Pmax | 650 W | Voc | 49.72 V |
| Vmp | 40.98 V | Isc | 16.54 A |
| Imp | 15.86 A | Module efficiency | up to 24.6% |
| Temp. coeff. Voc | −0.20 %/°C | Temp. coeff. Isc | +0.05 %/°C |
| Temp. coeff. Pmax | −0.26 %/°C | Max. system voltage | 1,500 V |
| Cells / glass | 132, dual glass, bifacial | Glass feature | Anti-glare, low reflection |

Design assumptions (Chiang Rai): coldest cell temperature 0 °C at dawn; hottest cell temperature 70 °C; Vmp temperature coefficient taken as −0.25 %/°C; bifacial rear-side gain +10% on a normal roof and +20% on a carport or light-coloured ground. Rear-gain values are design assumptions; confirm bifaciality with the LONGi datasheet.

## 3. String voltage check (same for both inverters)

| Panels per string | Voc @0 °C (limit 1,100 V) | Vmp @70 °C (MPPT min 200 / 180 V) | Vmp @25 °C | Result |
|---|---|---|---|---|
| 16 | 835 V | 582 V | 656 V | **OK** |
| 18 | 940 V | 655 V | 738 V | **OK** |
| 20 | 1,044 V | 727 V | 820 V | **OK** |
| 21 | 1,096 V | — | — | **Too tight, avoid** |

Use 16 to 20 panels per string. All strings are far above both start voltages, so the MC0's 180 V start gives no practical advantage with this panel.

## 4. String current check (where the models differ)

| Case | Value with this panel | 50KTL-M3 | 50K-MC0 |
|---|---|---|---|
| 1 string per input, design Isc (roof / carport) | 18.6 A / 20.3 A | Limit 20 A: **OK** roof / **marginal** carport | Limit 23 A: **OK** both |
| 2 strings per MPPT, Isc (roof / carport) | 37.2 A / 40.6 A | Limit 40 A: **OK** roof / **exceeds** carport | MPPT 2 & 3, limit 44 A: **OK** both |
| 2 strings per MPPT, operating current (STC / +10% rear) | 31.7 A / 34.9 A | Limit 30 A: **clips even at STC** | MPPT 2 & 3, limit 33 A: **slight clipping** with rear gain |

Design Isc = Isc × rear-side gain × hot-temperature factor (+2.25% at 70 °C). On the M3, two-string MPPTs lose a small amount of yield to current limiting (estimated under ~1%/year on those MPPTs); this is a yield issue, not a safety issue. On carports and high-albedo ground mounts the M3 can exceed its 40 A short-circuit limit, so use one string per MPPT there.

## 5. Recommended layouts

| Option | Panels | DC power | DC/AC ratio | Strings on MPPT 1 / 2 / 3 / 4 | Suitable for |
|---|---|---|---|---|---|
| A – Balanced | 96 (6 × 16) | 62.4 kWp | 1.25 | 1 / 2 / 2 / 1 | MC0: all sites. M3: rooftop |
| B – Extra morning/evening yield | 108 (6 × 18) | 70.2 kWp | 1.40 | 1 / 2 / 2 / 1 | MC0: all sites. M3: rooftop (more clipping) |
| C – M3 on carport / ground | 80 (4 × 20) | 52.0 kWp | 1.04 | 1 / 1 / 1 / 1 | M3 where rear gain is high |

On the MC0, place the two-string MPPTs on MPPT 2 and 3 (33 A / 44 A).

## 6. Recommendation

For new projects with the LONGi LR8-66HVD-650M anti-glare panel, **choose the SUN2000-50K-MC0** if our supplier can deliver it at a similar price, especially for carports and ground mounts. The **SUN2000-50KTL-M3** remains a sound choice for rooftop jobs if it is noticeably cheaper or already in stock, accepting slight clipping on two-string MPPTs.

**Before quoting the MC0:** confirm it is on PEA's approved grid-connected inverter list. As a new model, it may not be listed yet, which would delay the grid-connection application.

## Sources

- Huawei – SUN2000-50KTL-M3 technical specification (solar.huawei.com/th/products/SUN2000-50KTL-M3/specs)
- Huawei – SUN2000-50K-MC0 datasheet, version 01-202510
- LONGi LR8-66HVD-650M specifications (sun.store product listing); anti-glare feature (HomePro Thailand listing)

Figures are based on manufacturer datasheets and the design assumptions stated above. Final string design should be confirmed against site conditions and the latest datasheets.

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